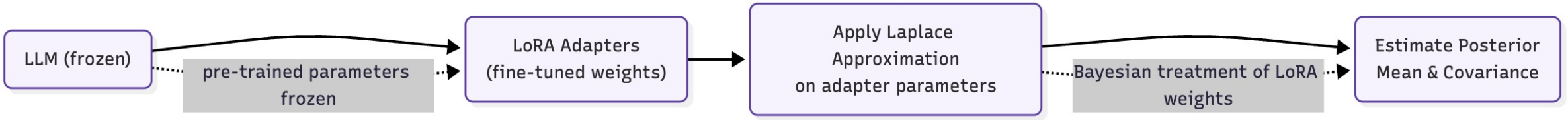
Laplace-LoRA

Laplace-LoRA[115]

☐ Core Idea

- Fine-tuned LLMs often become overconfident, especially when trained on small datasets. Laplace-LoRA introduces a Bayesian perspective to LoRA fine-tuning by approximating the epistemic uncertainty over LoRA parameters using a Laplace approximation.



[115] Bayesian Low-rank Adaptation for Large Language Models, arxiv 2023

Laplace-LoRA[115]

- **Task Type:** Multiple-choice QA & Binary reasoning

- **Introduction:** Laplace-LoRA applies a Laplace approximation to the low-rank adapter parameters after fine-tuning, introducing a Bayesian posterior over LoRA updates while keeping the backbone fixed. It quantifies parameter uncertainty and provides a distributional view of LoRA-based adaptation within standard LLM fine-tuning workflows.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Bayesian	Parameter	White-box	Post-hoc	High	Ranking/Calibration

[115] Bayesian Low-rank Adaptation for Large Language Models, arxiv 2023